

Motor Synergy Generalization Framework: New Targets in Multi-Finger In-Hand Manipulation

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Motivation

In-hand manipulation

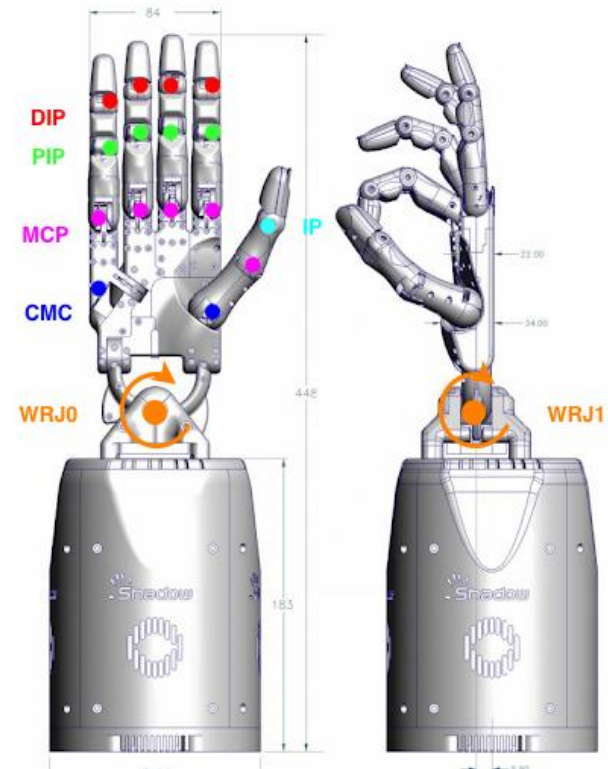
- Reorient an object, without putting it down
- Requires coordinated motion of fingers and contact forces, with sensory feedback
- Humans do it efficiently: **can robots learn to do the same?**



Reinforcement Learning (RL)

Problem of sampling inefficiency

- Recent works have demonstrated that RL can produce impressive dexterous manipulation behaviour
- Applying RL to anthropomorphic hands is challenging due to the **high dimensionality** of the action space (≥ 20 DoF)
- Standard RL is **sample-inefficient** in high dimensional continuous spaces

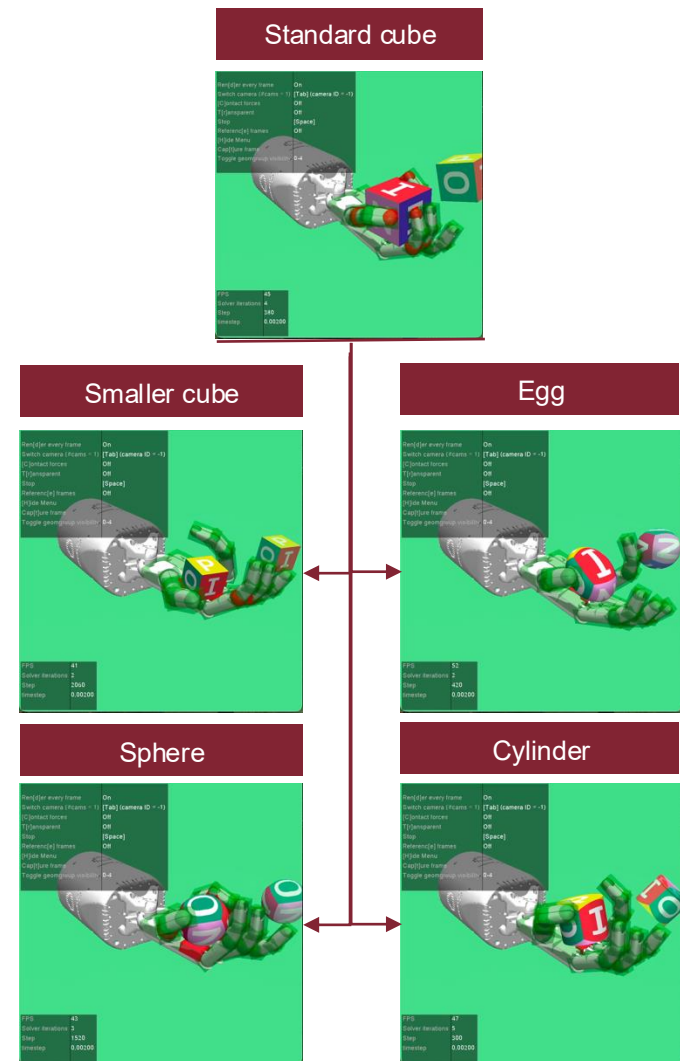


| The Shadow Dexterous Hand with 24 DoF

Solution

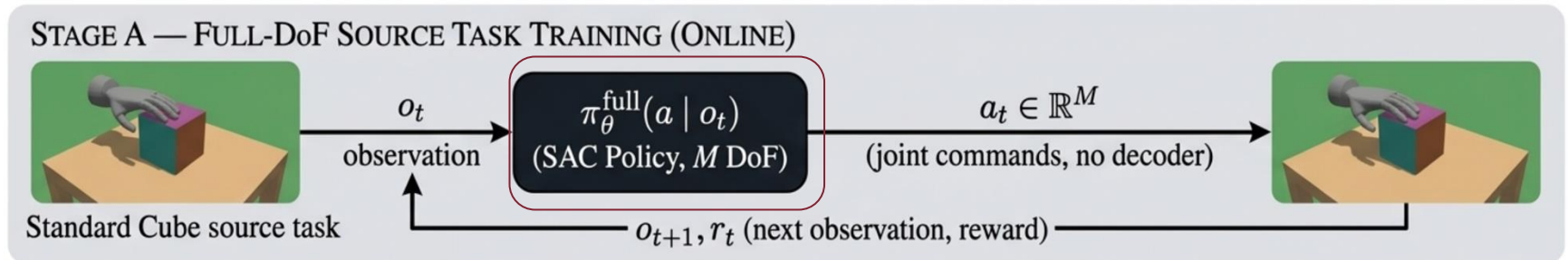
Motor synergies

- Human motor system manages complexity by organizing groups of muscles and joints into coordinated motion activation patterns:
motor synergies
- Searching in a smaller space of (one or more) synergy activations decreases drastically the dimensionality of the control problem
- Recent works have validated the transferability of motor synergies to new robotic tasks
- **Can synergies extracted from manipulation of one object be used to accelerate learning for in-hand manipulation of new objects?**



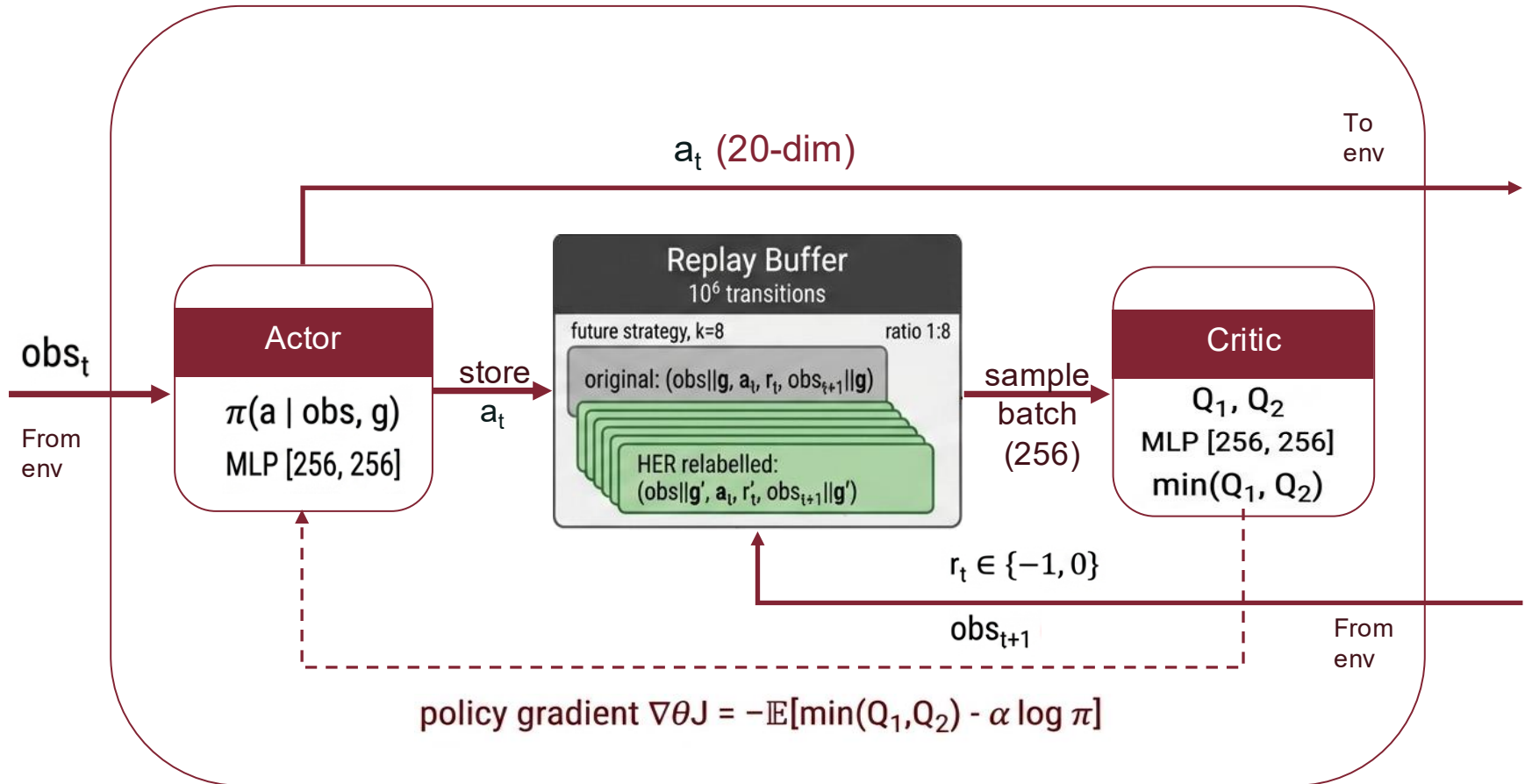
Proposed method

Motor Synergy Generalization Framework



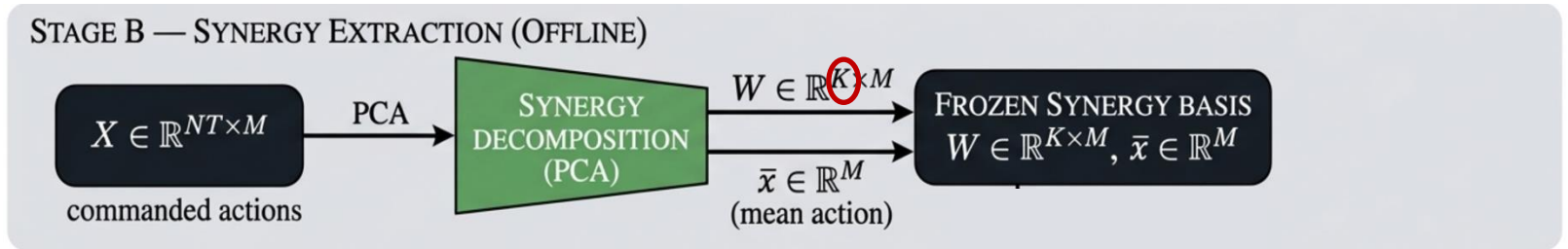
Proposed method

Stage A: Full-DoF Source Task Training



Proposed method

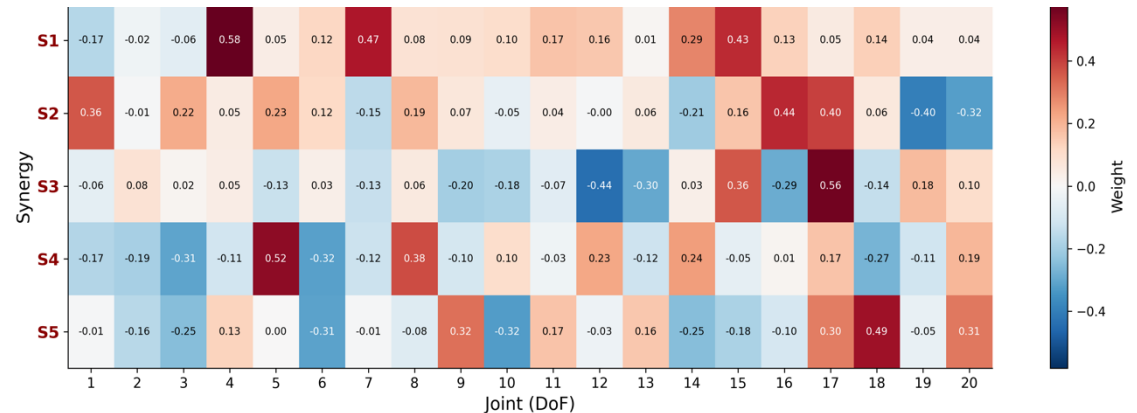
Stage B



DATA COLLECTION

- $M = 20$
- $N = 200$
- $T = 100$

W

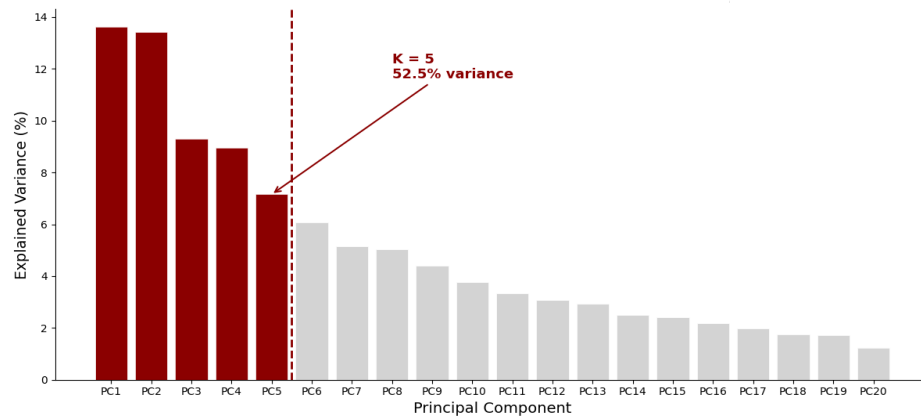
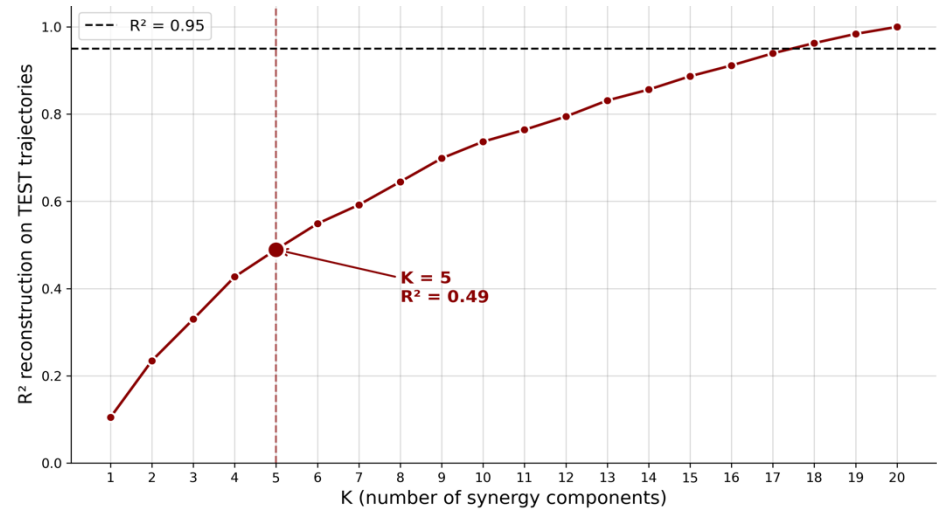


Proposed method

PCA: Why K=5 synergies?

Reconstruction performance

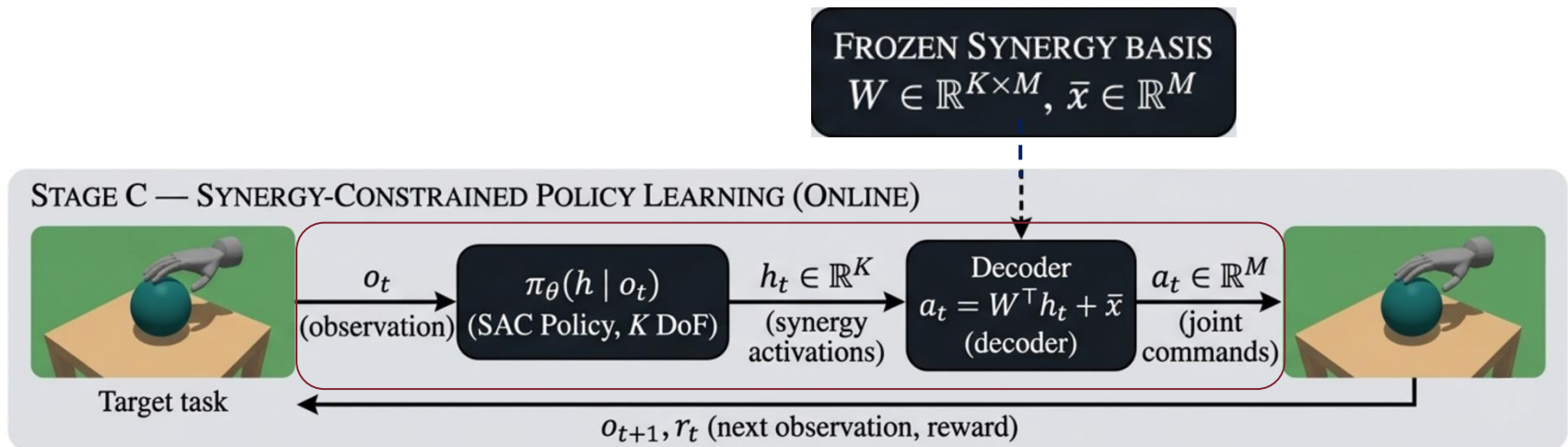
$$R^2(K) = 1 - \frac{\sum_i \|x_i - \hat{x}_i^{(K)}\|^2}{\sum_i \|x_i - \bar{x}\|^2}$$



Variance explained by each synergy component

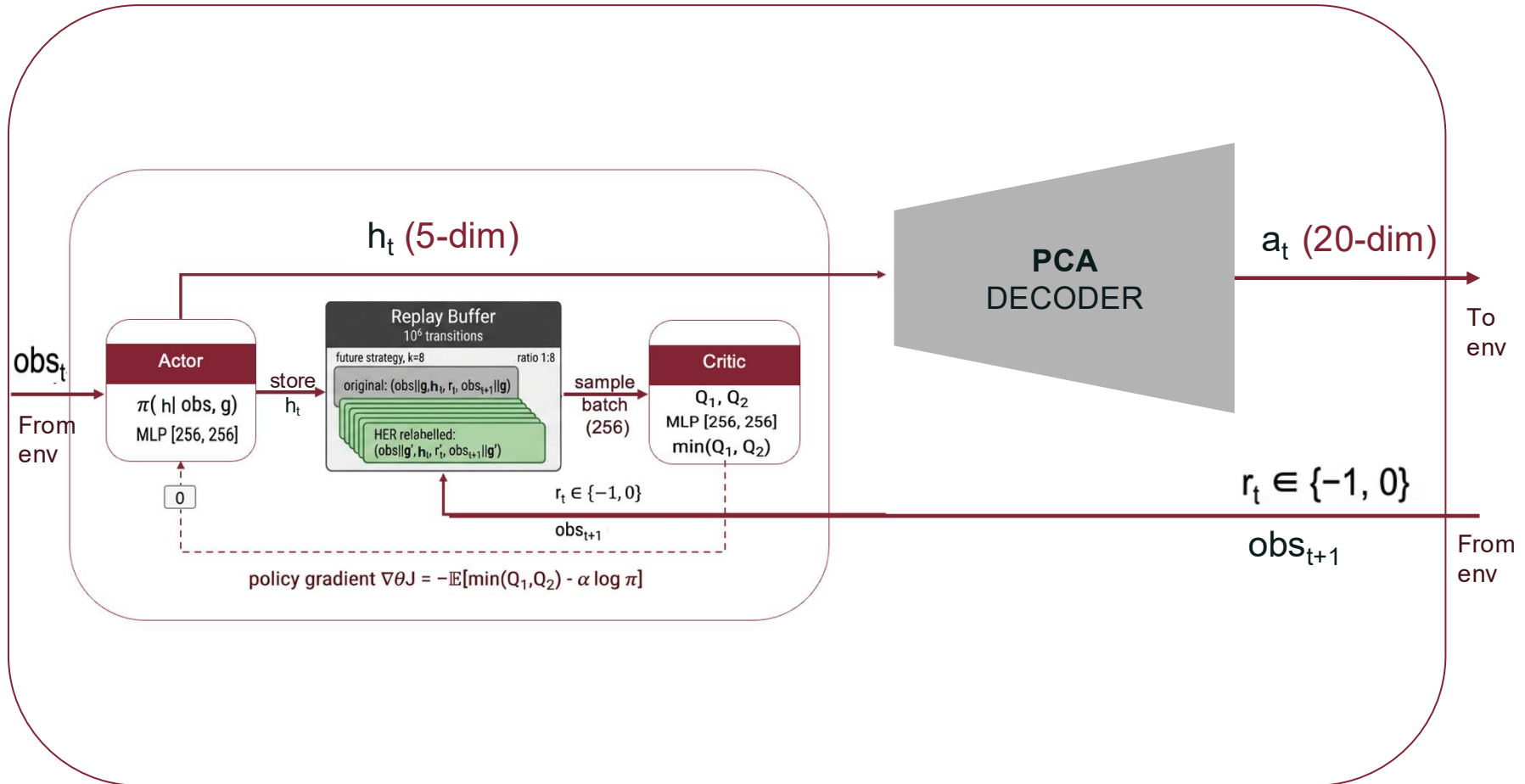
Proposed method

Stage C



Proposed Method

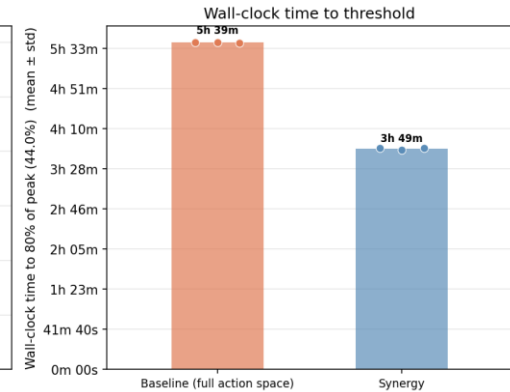
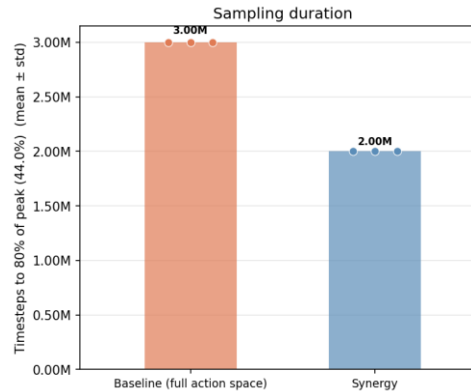
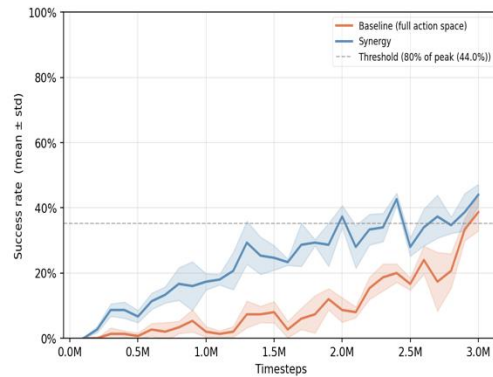
Stage C: Synergy-Constrained Policy Learning



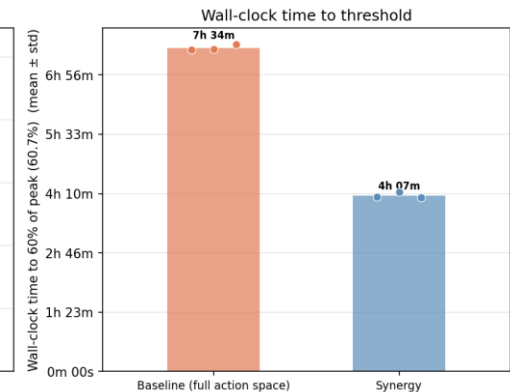
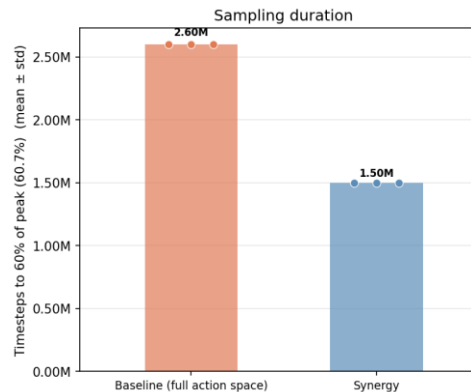
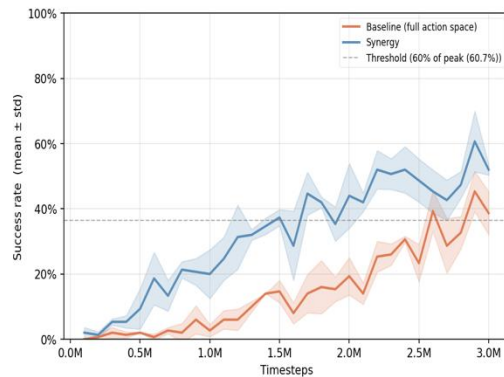
Results

Target tasks: Cube variants

Large cube



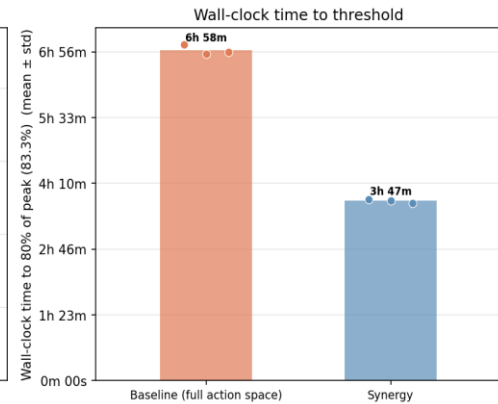
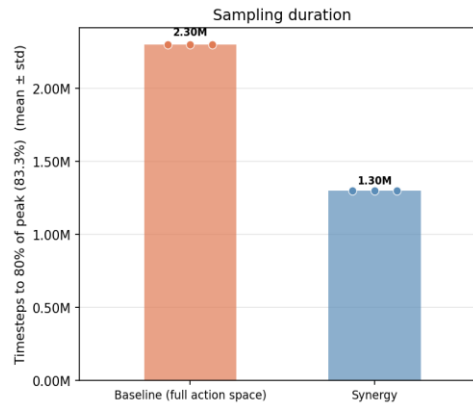
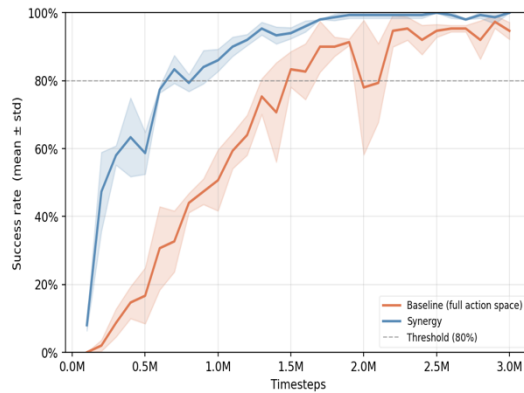
Small cube



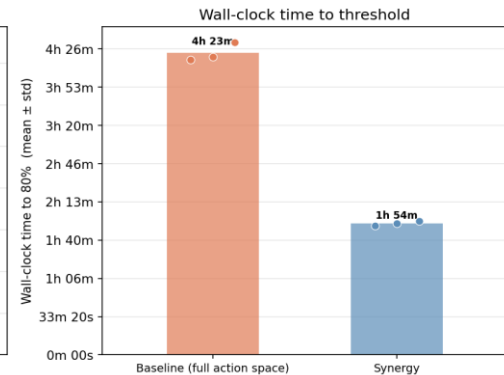
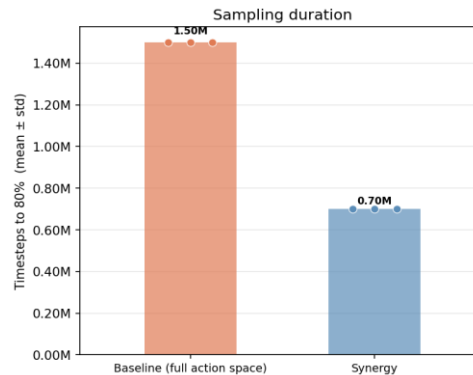
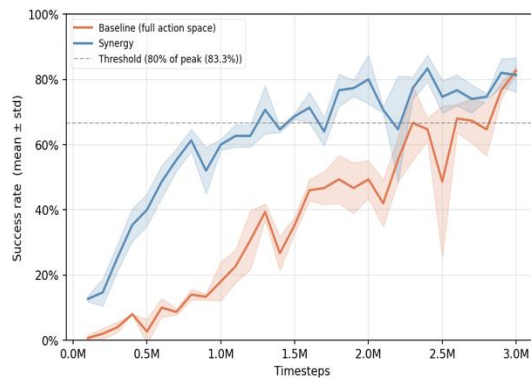
Results

Target tasks: Egg and cylinder

Egg



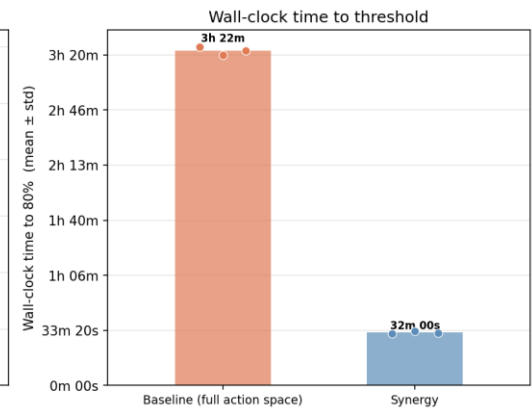
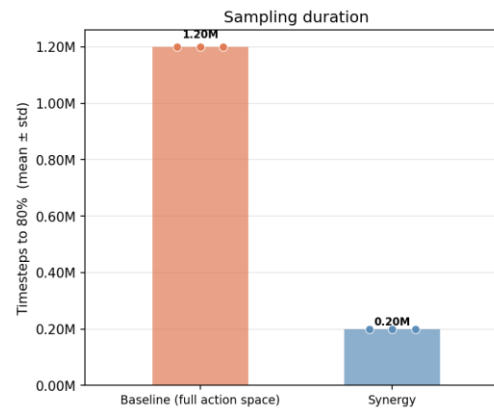
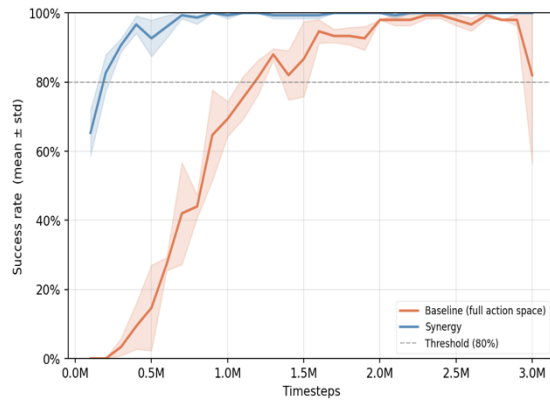
Cylinder



Results

Target tasks: Sphere

Sphere



Results

Quantitative results

Target Object	Training Steps ($\times 10^6$)		Wall-Clock Time (h)		Speedup
	Full 20 DoF	Synergy 5 DoF	Full 20 DoF	Synergy 5 DoF	
Large Cube	3	2	5.65	3.82	~ 1.5 ×
Small Cube	2.6	1.5	7.57	4.12	~ 1.8 ×
Egg	2.3	1.3	6.97	3.78	~ 1.8 ×
Cylinder	1.5	0.7	4.38	1.9	~ 2.3 ×
Sphere	1.2	0.2	3.37	0.53	~ 6.4 ×

Conclusions and future work

➤ Conclusions

Constraining the action space to the synergy subspace extracted from a single source task reduces learning time

The sampling duration and wall-clock time required to reach a given level of performance reduce considerably across all target objects

This result is consistent regardless of the object geometry

➤ Future work

Evaluation without tactile sensing

Alternative synergy extraction methods

Larger and more diverse object sets

Multi-source synergy extraction

Sim-to-real transfer